BS(Artificial Intelligence)
Fall-2024
Computer Organization & Assembly Language

Course Title: Computer Organization & Assembly Language
Course Code: CSC-324
Credit Hours: (2+1)
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Description:

The theory component of CSC-324 covers the organization and architecture of digital computers — CPU structure, memory hierarchy, I/O organization, instruction set design, addressing modes, digital logic, computer arithmetic, and control unit operation. It builds a conceptual foundation for how computers execute instructions at every level, from the fetch-decode-execute cycle up to pipelining, caching, and parallel/multicore organization, preparing students to reason about performance and design trade-offs rather than just memorize architectural facts.

Aims and Objectives:

1. To explain the structure and function of the CPU, memory, and I/O subsystems and how they interconnect within a computer system.
2. To analyze instruction set architectures, addressing modes, and assembly-level program behavior across different processor designs (CISC and RISC).
3. To describe digital logic fundamentals, computer arithmetic, and control unit design (hardwired and microprogrammed) as the basis for processor implementation.
4. To evaluate architectural performance trade-offs — including pipelining, cache design, and multicore/parallel organization — in modern processor design.

Assessment:

S. No	**Assessment Activities**	Percentage	Total Activities
1.	Sessional: Quizzes/ Assignments (Quizzes & Assignments)	30%	Multiple
2.	Mid Term Exam	30%	1
3.	Final Exam	40%	1

Course Learning Outcomes (CLOs):

No.	**Course Learning Outcome**	Domain	GA	Assessment Tool
C1	**Understand** the organization and architecture of digital computers, including CPU, memory, I/O, and instruction set structures.	C2	GA2	Class Participation, Quizzes, Mid Exams., Assignments, Final Exam
C2	**Demonstrate** assembly language programming concepts and RISC-V instruction execution using industry-standard simulation tools.	C3	GA3	Class Activity, Quiz, Assignments, Mid Exam, Final Exam
C3	**Design and implement** digital logic circuits and a simple processor datapath to solve computational problems.	C3	GA4	Worksheets, Project

Domains:
C=Cognitive, A=Affective, P=Psychomotor0
Levels:
Cognitive = {1: Remembering, 2: Understanding, 3: Applying, 4: Analyzing, 5: Evaluating, 5: Creating}
Affective = {1: Receiving, 2: Responding, 3: Valuing, 4: Organizing, 5: Characterizing}
Psychomotor= {1: Imitation, 2: Manipulation, 3: Precision, 4: Articulation, 5: Naturalization}

Course Outlines:

Week	Lec#	Subtopics	Reference	Course % Covered
Week 01	Lec 01	1.1 Organization vs. Architecture; Structure & Function	Ch. 1	3.13%
	Lec 02	1.2 The IAS Computer; x86 Evolution; Embedded Systems & ARM	Ch. 1	6.25%
Week 02	Lec 03	Flags & Basic Arithmetic Primer (ZF/CF/OF/SF)	Ch. 11 (intro)	9.38%
	Lec 04	8.1 External Devices, I/O Modules	Ch. 8	12.50%
Week 03	Lec 05	8.2 Programmed I/O, Interrupt-Driven I/O	Ch. 8	15.63%
	Lec 06	3.1 Computer Components, Interconnection, Bus Structures	Ch. 3	18.75%
Week 04	Lec 07	3.2 Bus Arbitration	Ch. 3	21.88%
	Lec 08	13.1 Instruction Set: Operand Types	Ch. 13	25.00%
Week 05	Lec 09	13.2 Instruction Set: Operation Types	Ch. 13	28.13%
	Lec 10	14.1 Addressing Modes	Ch. 14	31.25%
Week 06	Lec 11	14.2 Instruction Formats	Ch. 14	34.38%
	Lec 12	15.1 Assembly Language Concepts, Assemblers	Ch. 15	37.50%
Week 07	Lec 13	15.2 Loading & Linking + Midterm Review	Ch. 15	40.63%
	Lec 14	16.1 Processor & Register Organization, Instruction Cycle	Ch. 16	43.75%
Week 08	Lec 15	16.2 Single-Cycle Instruction Execution	Ch. 16	46.88%
	Lec 16	16.3 Instruction Pipelining Basics	Ch. 16	50.00%
Week 09	Lec 17	16.4 Pipeline Hazards (structural, data, control)	Ch. 16	53.13%
	Lec 18	4.1 Memory Hierarchy, Locality of Reference	Ch. 4	56.25%
Week 10	Lec 19	5.1 Cache Principles & Mapping Techniques	Ch. 5	59.38%
	Lec 20	17.1 RISC vs. CISC Characteristics	Ch. 17	62.50%
Week 11	Lec 21	RISC-V in Industry — open ISA, adoption, custom silicon	Industry-curated	65.63%
	Lec 22	2.1 Performance Metrics, Amdahl's Law	Ch. 2	68.75%
Week	Lec	12.1 Boolean Algebra, Logic Gates	Ch. 12	71.88%

12	23			
	Lec 24	12.2 Half/Full Adders, Basic Combinational Logic	Ch. 12	75.00%
Week 13	Lec 25	12.2 Combinational Circuits: Multiplexers, Decoders	Ch. 12	78.13%
	Lec 26	12.2 ALU Design	Ch. 12	81.25%
Week 14	Lec 27	12.3 Sequential Circuits: Flip-Flops, Registers, PLDs	Ch. 12	84.38%
	Lec 28	11.1 Integer Arithmetic — Two's Complement, Overflow (full)	Ch. 11	87.50%
Week 15	Lec 29	11.2 Floating-Point Arithmetic (IEEE 754)	Ch. 11	90.63%
	Lec 30	16.5 Datapath-Level Register Organization	Ch. 16	93.75%
Week 16	Lec 31	19.1/19.2 Control Unit: Hardwired & Microprogrammed Control	Ch. 19	96.88%
	Lec 32	20/21 Parallel Organization (Multicore, MESI) + Course Review	Ch. 20/21	100%

Textbooks

1. Computer Organization and Architecture: Designing for Performance, 11th Editionby William Stallings
2. Computer Architecture: A Quantitative Approach — Hennessy & Patterson

Reference Books:

1. Digital Design and Computer Architecture, RISC-V Edition — Sarah & David Harris
2. Computer Organization and Design: The Hardware/Software Interface, RISC-V Edition — Patterson & Hennessy

Prepared by_____ Verified by_____

HOD_____